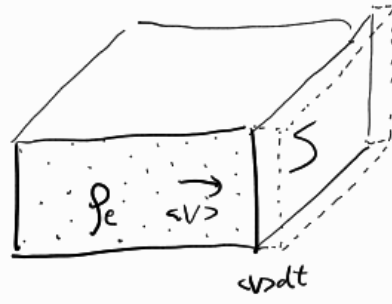


premikanje nabojov - tok

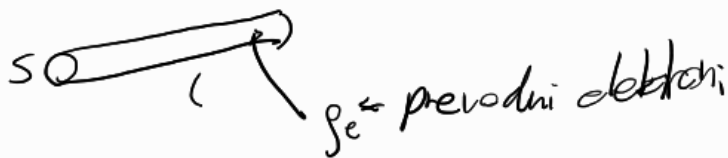
$$I = \frac{de}{dt} = [A]$$



$$dV = \langle v \rangle dt \cdot S$$

$$I = \frac{\rho_e dV}{dt} = \rho_e S \langle v \rangle$$

žica

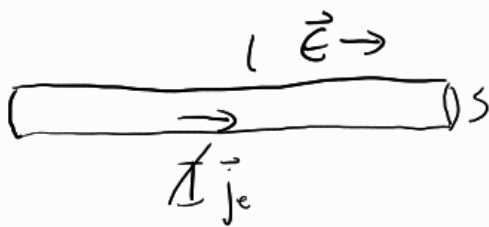


$$\rho_e = n e_0 = \rho \frac{N_A}{M} e_0 \quad (\times 1) \quad \varphi$$

št. prostih elektronov

$$j_e = \frac{dI}{dS} = \rho_e \langle v \rangle ; \quad \vec{j}_e = \rho_e \langle \vec{v} \rangle$$

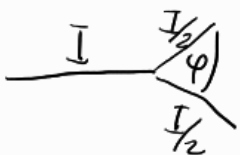
ρ
gostota toka



homogen $\vec{j} \rightarrow$ homogen $\vec{E}$

$$U = EI$$

I ni vektor!



Tak se razdeli na pol neodvisno od izbire φ

hitrost e^- v lni (žica = baker)

$$I = 1A$$

$$S = 1 \text{ mm}^2$$

$$\langle v \rangle = \frac{I/S}{\rho_e} = \frac{1A \cdot m^2 \cdot 100 \text{ kg}}{1 \text{ mm}^2 \cdot 10^{-6} \text{ kg} \cdot 6 \cdot 10^{26} \text{ A}^{-1}} = 10^{-4} \frac{m}{s}$$

$$v_{\text{term}} \approx 10^6 \text{ m/s}$$

} podobno kot pri zvoku



Ohmov zakon

$[U \propto I] \rightarrow V = RI$
 $\uparrow$ upor $[\frac{V}{A} = \Omega]$
 $\uparrow$ Ohm
 $\downarrow$ v žici: $E = \frac{U}{l}$ $j = \frac{1}{S}$

$$\epsilon l = R j \epsilon S$$

$$\vec{j} = \underbrace{\frac{l}{RS}}_{\text{(specifična) prevodnost}} \vec{E}$$

$$\sigma = \frac{l}{RS}$$

$\rightarrow$ specifični upor $\frac{1}{\sigma} = \underbrace{\rho = \frac{RS}{l}}_{\text{specifični upor}} [\Omega m, \Omega \frac{mm^2}{m}]$

$$R_{\text{žice}} = \frac{\rho l}{S}$$

material	$\rho [\Omega m]$
superprevodniki	0
Ag	$1,62 \cdot 10^{-8}$
Cu	$1,7 \cdot 10^{-8}$
Al	$2,75 \cdot 10^{-8}$
diamant	10^{16}

$\uparrow$ prevodniki
 $\downarrow$ izolatorji

$$\vec{E} = \sum \vec{j}$$

$$\langle \vec{v} \rangle = \vec{a} t = \frac{F_e}{m_e} t = - \frac{e_0 E}{m_e} t$$

Drude $\left\{ \begin{array}{l} \rightarrow \text{trki z ioni v kristalni mreži } m_{ion} \gg m_e \\ \text{po trku } v \approx v' \\ \langle \vec{v} \rangle \approx 0 \text{ (vse smeri)} \end{array} \right.$

v_x vs t graph showing a sawtooth pattern. $\frac{dv}{dl} \propto e^{-\frac{l}{\langle l \rangle}}$ $\frac{dv}{dt} = e^{-\frac{t}{\tau}}$ $\langle l \rangle = \langle v \rangle \tau$

$\rightarrow \langle v_x \rangle = a \tau = \frac{e_0 E}{m_e} \tau$

$j = e_0 n_e \langle v_x \rangle = \frac{e_0^2 \rho N_A \tau}{m_e} E$

(termična lastnost) $\langle v_x \rangle \ll \langle v \rangle$
 "kost", ker zaradi upliva e hitrost e

za Cu:

$\tau = \frac{m_e}{e_0^2 \rho N_A} = \frac{10^{-30} \text{ kg} \cdot (3,5 \text{ kg m}^{-3})^{-1} \cdot 10^9 \text{ S}}{2,5 \cdot 10^{-28} \text{ A}^2 \cdot 8960 \text{ kg} \cdot 6 \cdot 10^{26} \cdot 1,7 \cdot 10^{-8} \text{ Ohm m}} = 2,8 \cdot 10^{-14} \text{ s}$

$VA_s = j = Nm$

$\underline{2,8 \cdot 10^{-14} \text{ s}}$

$\langle l \rangle = \langle v \rangle \tau \approx 30 \text{ nm}$

$T \uparrow \Rightarrow \tau \downarrow \Rightarrow \xi \uparrow$

$\frac{dR}{R} = \frac{d\xi}{\xi} = \alpha dT$

	$\alpha [K^{-1}]$
Ag	4,3
Cu	4,3
Al	4,4
Pt	3,9

$\cdot 10^{-3}$

$\frac{\Delta R}{R} = \alpha \Delta T (+ \beta \Delta T^2 + \gamma \Delta T^3 + \dots)$

Polprevodnik

Si, Ge

	Si (čist)	Cu
$\rho [\Omega m]$	$3 \cdot 10^3$	$2 \cdot 10^{-8}$
$\alpha [K^{-1}]$	$7 \cdot 10^{-3}$	$4 \cdot 10^{-3}$
$n [m^{-3}]$	10^{16}	10^{29}

$\xi \propto \frac{1}{n_e \tau}$
 $T_{Si} \approx T_{Cu}$
 $\Downarrow$
 $n_{Si} \ll n_{Cu}$

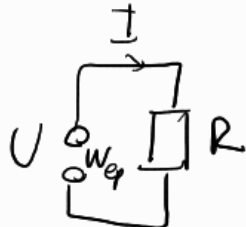
1911 - Onnes:

$Hg, T < 4K \rightarrow \underline{\underline{\xi(Hg) \equiv 0}}$

Superprevodnik

- keramike so superprevodniki že pri $T > 77K$

Moč



$W_{ep} = eU$

$P = \frac{dA_e}{dt} = UI \quad [W = VA]$

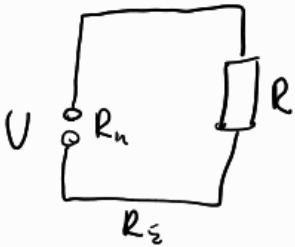
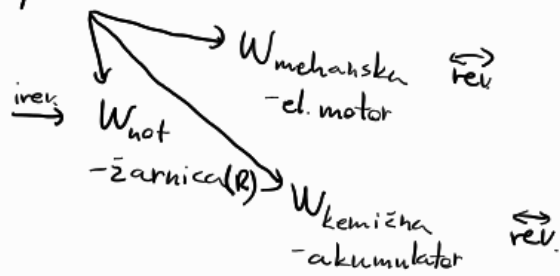
$dA_e = dW_{ep} = Ude = UI dt$

$P = UI = I^2 R = \frac{U^2}{R}$

baterija vs. kondenzator



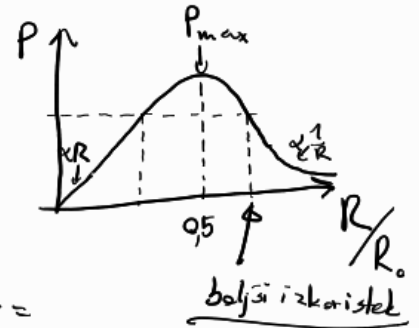
$$-dW_{cp} = dW_{porabnik}$$



$$P = UI = \frac{U^2}{R}$$

$$P_{max} = ?$$

$$\bar{I} = \frac{U}{R + R_0}; P_R = \bar{I}^2 R = U^2 \frac{R}{(R + R_0)^2}$$

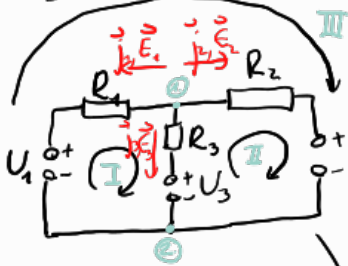


$$\eta = \frac{I^2 R}{UI} = \frac{UR}{U(R + R_0)} = \frac{R}{R + R_0}$$

$$\frac{dP_R}{dR} = 0 = \frac{(R + R_0)^2 - R(2(R + R_0))}{...}$$

$$= (R + R_0)(R + R_0 - 2R) \rightarrow \underline{\underline{R = R_0}}$$

krog z več zankami

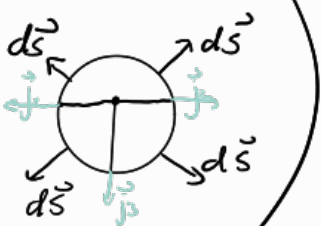


$$\oint \vec{E} \cdot d\vec{s} = 0$$

$$I = \int \vec{j} \cdot d\vec{s}$$

$$U = -\int \vec{E} \cdot d\vec{s}$$

$$\vec{j} = \frac{1}{\epsilon} \vec{E}$$



$$\sum U_i = 0$$

baterije uporabi

$$-d\vec{s} \text{ v smeri } \vec{j} \rightarrow -RI$$

$$-d\vec{s} \text{ nasproti } \vec{j} \rightarrow RI$$

$$-d\vec{s} \text{ iz } + \rightarrow - \Rightarrow -U_j$$

$$-d\vec{s} \text{ iz } - \rightarrow + \Rightarrow +U_j$$

"pravi smer"

$$\oint j_1 dS_1 + \oint j_2 dS_2 + \oint j_3 dS_3 = 0$$

$$I_1 + I_2 + I_3 = 0$$

$$-I_1 - I_2 - I_3 = 0$$

1. Kirchhoffov zakon:

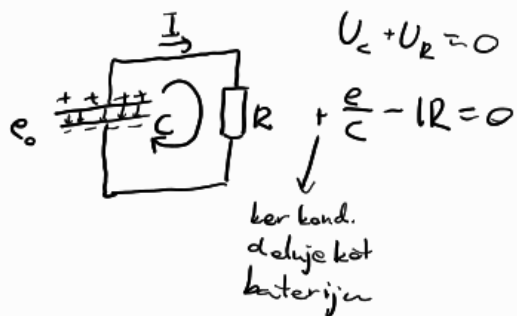
- vsota tokov v razvejišču je 0 (če $\vec{j}$ kaže ven iz zaklj. površine, je I pozitiven, če $\vec{j}$ kaže v zaklj. površino je I negativen)

2. Kirchhoffov zakon:

- vsota napetosti v zaključeni zanki je 0

Kondenzator

1) Praznjenje



$$I = -\frac{de}{dt}$$

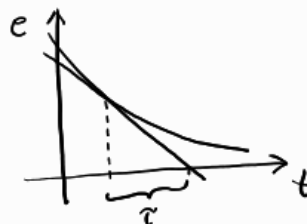
$$\frac{e}{C} = -\frac{de}{dt} R$$

$$\int_{e_0}^e \frac{de}{e} = \int_0^t \frac{1}{RC} dt$$

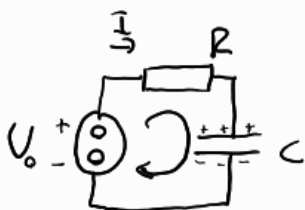
$$\ln \frac{e}{e_0} = -\frac{t}{\tau}$$

$$\boxed{e = e_0 e^{-\frac{t}{\tau}}}$$

$$\tau = RC$$



2) Polnjenje



$$U_0 - RI - \frac{e}{C} = 0$$

$$e_0 - \frac{de}{dt} RC - e = 0$$

$$e_0 - e = \frac{de}{dt} RC$$

$$\int_0^t \frac{1}{\tau} dt = \int_{e_0-e}^e \frac{de}{e_0-e}$$

$$-\frac{t}{\tau} = \ln \frac{e_0 - e}{e_0}$$

$$e_0 - e = e_0 e^{-\frac{t}{\tau}}$$

$$e = e_0 (1 - e^{-\frac{t}{\tau}})$$

$$\eta = \frac{W_c}{A_e} = \frac{1}{2}$$

boljši izkoristek, če napetost počasi večamo

